Operating Systems

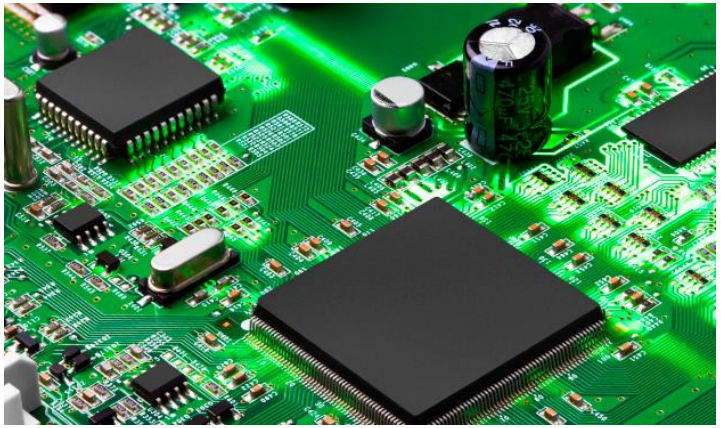
Chapter 1 Fundamentals

Tien Pham Van, Dr. rer. nat.

Compiled with reference to other presentations

Introduction

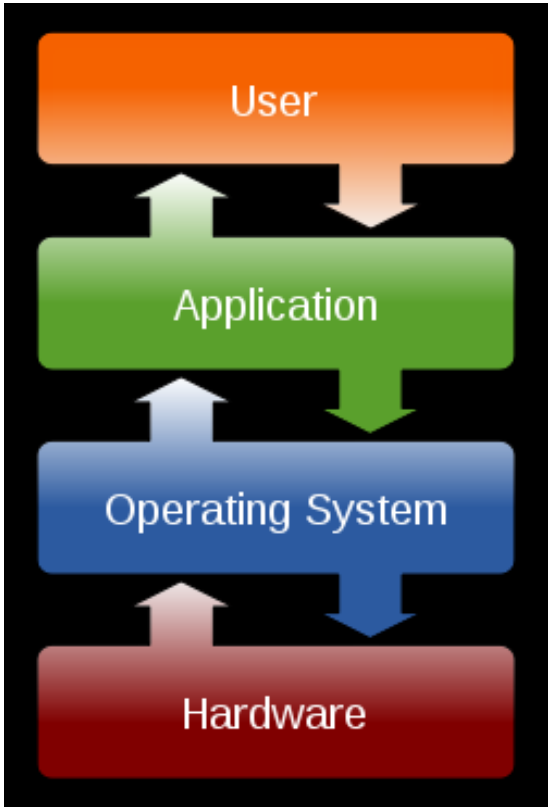
Banking system	Airline reservation	Web browser	Application programs
Compilers	Editors	Command interpreter	System programs
Operating system			
Machine language			Hardware
Microarchitecture			
Physical devices			



- A computer system consists of
  - hardware
  - system programs
  - application programs

Operating System

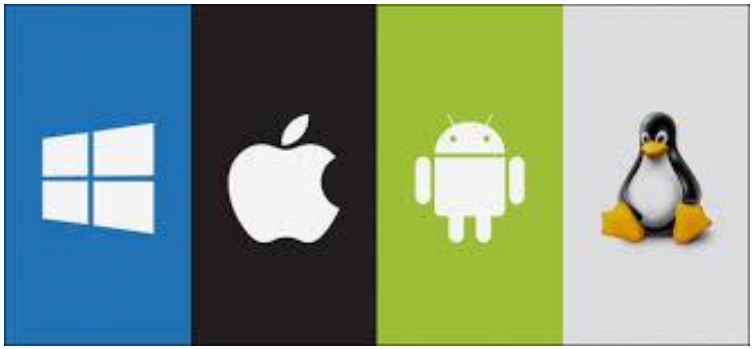
- A program that controls the execution of application programs
- An interface between applications and hardware



What is an Operating System

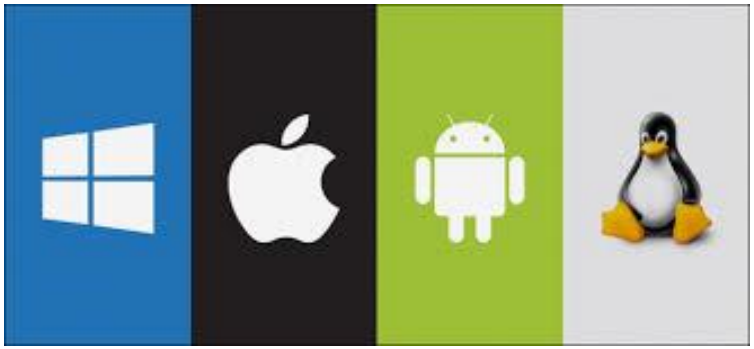
- It is an extended machine
  - Hides the messy details which must be performed
  - Presents user with a virtual machine, easier to use

- It is a resource manager
  - Each program gets time with the resource
  - Each program gets space on the resource



Then when does a computer need OS?

- Programs?
- Users/admins?
- System developments?
- Hardware control?



Performance Criteria

- Convenience
  - Makes the computer more convenient to use
- Efficiency
  - Allows computer system resources to be used in an efficient manner
- Ability to evolve
  - Permit effective development, testing, and introduction of new system functions without interfering with service

History of Operating Systems (1/3)

- **1940s–1950s: No OS (Manual Operation)**
  - Programs loaded manually using switches or punch cards
  - One job at a time
  - No abstraction or automation
- **1960s: Batch Systems**
  - Jobs grouped and executed sequentially
  - Introduction of **resident monitors**
  - Early mainframe operating systems (IBM OS/360)

History of Operating Systems (2/3)

- **1970s: Multiprogramming & Time-Sharing**
  - Multiple users interact simultaneously
  - CPU scheduling introduced
  - UNIX (1969–1973) → portability, C language
- **1980s: Personal Computer Era**
  - MS-DOS, Mac OS
  - Graphical User Interfaces (GUI)
  - Single-user systems become widespread

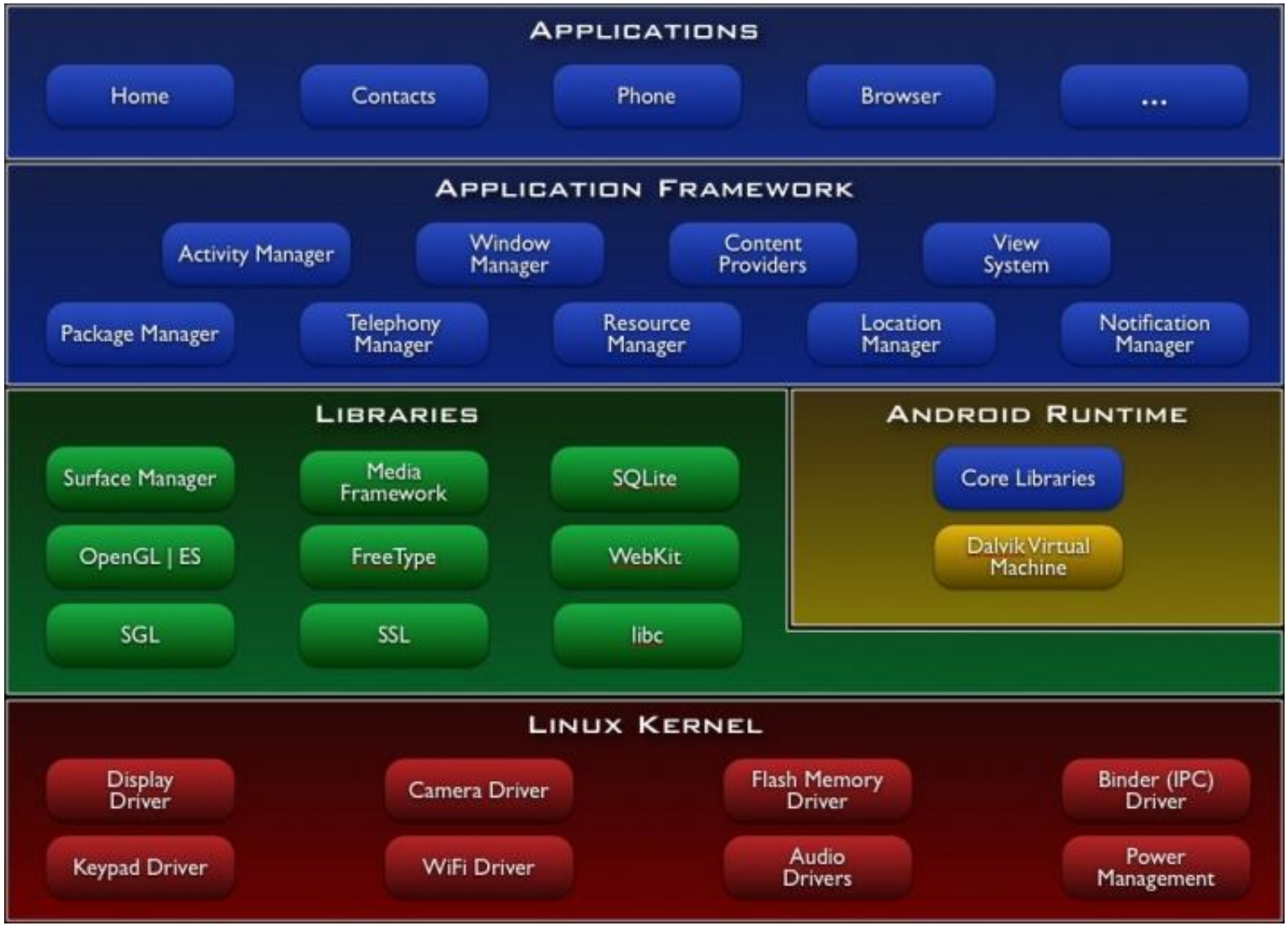
History of Operating Systems (3/3)

- **1990s: Networked & Distributed Systems**
  - Windows NT, Linux
  - Client–server computing
  - Internet-driven OS evolution
- **2000s–Present: Mobile, Virtualization & Cloud**
  - Android, iOS
  - Hypervisors & virtual machines
  - Cloud computing & containerization
  - Focus on security, scalability, and energy efficiency

Why study operating systems?

- Operating systems are a mature field
  - Most people use a handful of mature Oses
  - Hard to get people to switch operating systems
  - Hard to have impact with a new OS
- Still open questions in operating systems
  - Security– Hard to achieve security without a solid foundation
  - Scalability– How to adapt concepts when hardware scales 10×
- High-performance servers are an OS issue
  - Face many of the same issues as OSes, sometimes bypass OS
- Resource consumption is an OS issue
  - Battery life, radio spectrum, etc.
- New “smart” devices need new OSes

Modern OS


APPLICATIONS
Home
Contacts
Phone
Browser
...
APPLICATION FRAMEWORK
Activity Manager
Window Manager
Content Providers
View System
Package Manager
Telephony Manager
Resource Manager
Location Manager
Notification Manager
LIBRARIES
Surface Manager
Media Framework
SQLite
OpenGL | ES
FreeType
WebKit
SGL
SSL
libc
ANDROID RUNTIME
Core Libraries
Dalvik Virtual Machine
LINUX KERNEL
Display Driver
Camera Driver
Flash Memory Driver
Binder (IPC) Driver
Keypad Driver
WiFi Driver
Audio Drivers
Power Management

The Operating System family

- Mainframe operating systems
- Server operating systems
- Distributed operating systems
- Personal computer operating systems
- Real-time operating systems
- Embedded operating systems





Popular OSs

- https://youtu.be/ChQ18B1hofI

Layers of Computer System

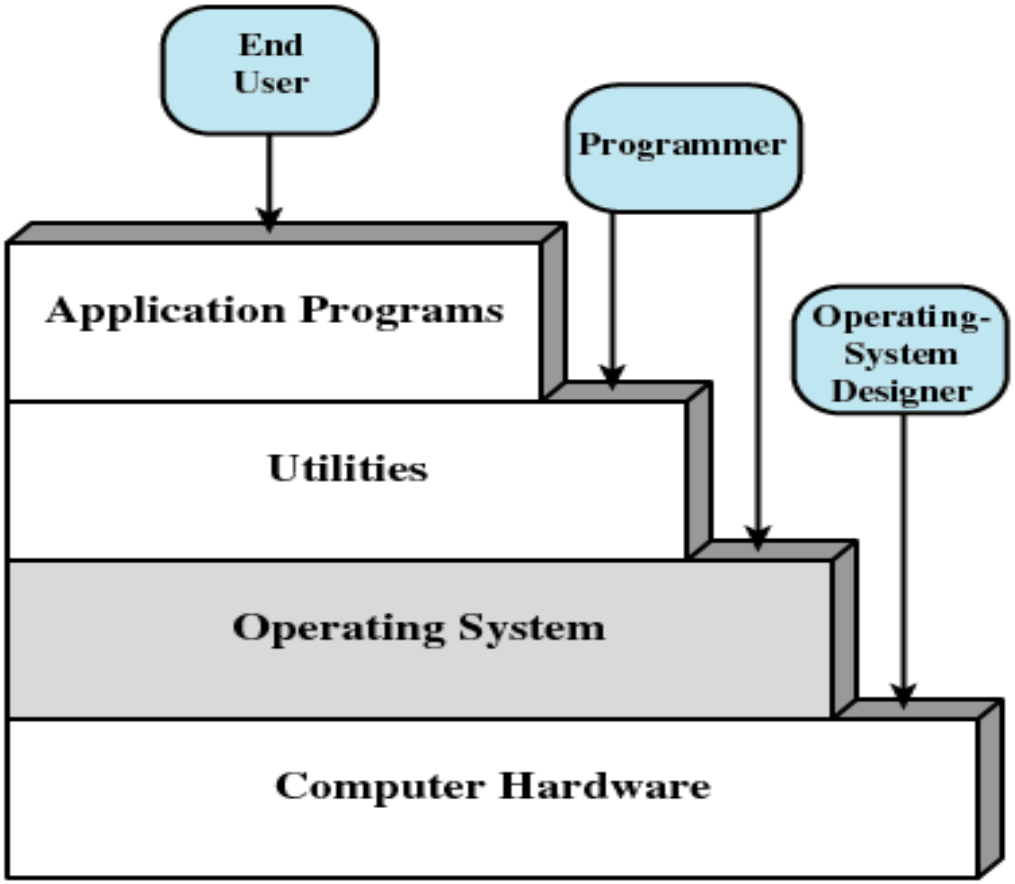


Figure 2.1 Layers and Views of a Computer System

Services Provided by the Operating System (1/3)

- Program development
  - Editors and debuggers
- Program execution
- Access to I/O devices
- Controlled access to files
- System access

Services Provided by the Operating System (2/3)

- Error detection and response
  - Internal and external hardware errors
    - Memory error
    - Device failure
  - Software errors
    - Arithmetic overflow
    - Access forbidden memory locations
  - Operating system cannot grant request of application

Services Provided by the Operating System (3/3)

- Accounting
  - Collect usage statistics
  - Monitor performance
  - Used to anticipate future enhancements
  - Used for billing purposes

Computers under OS's control

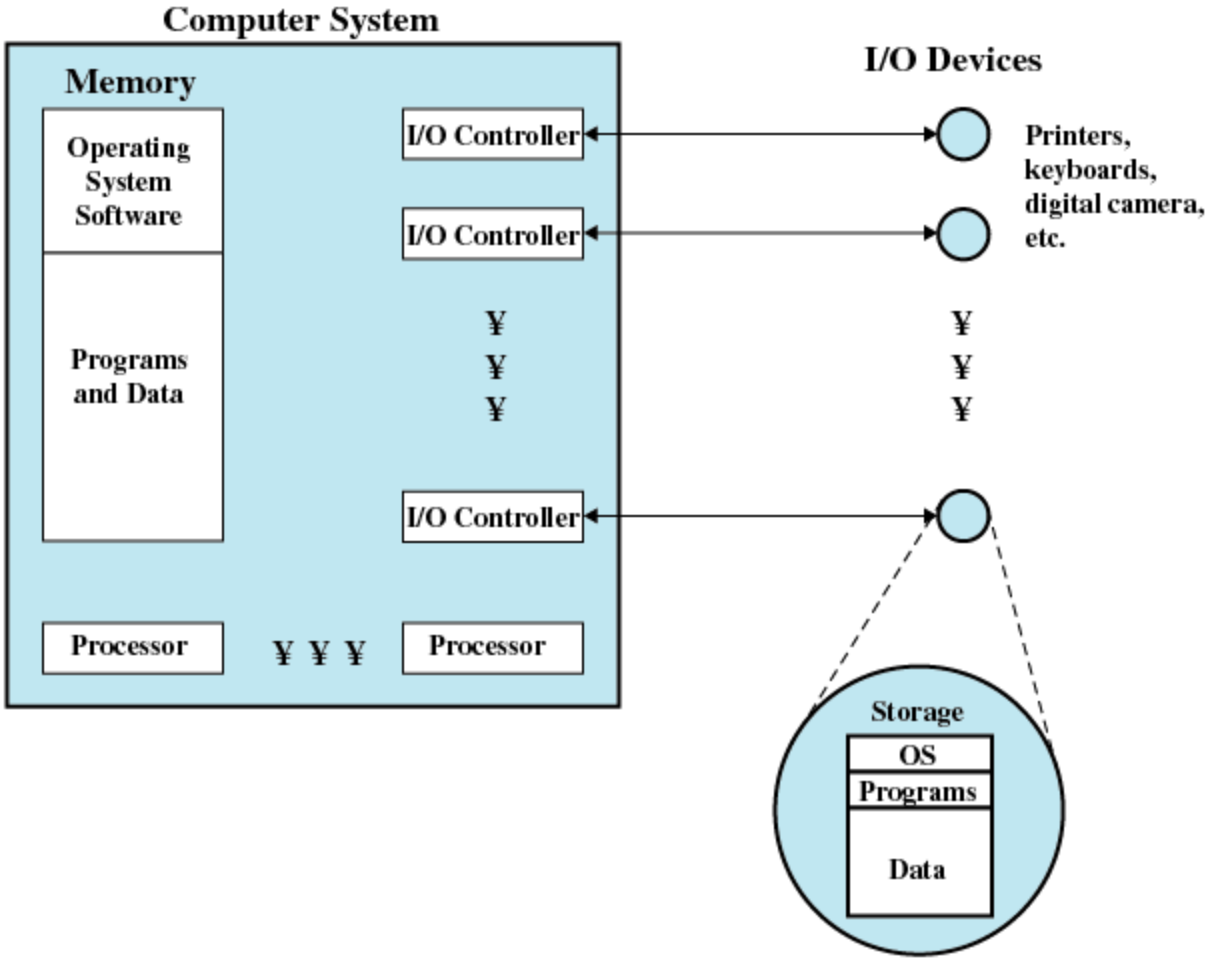
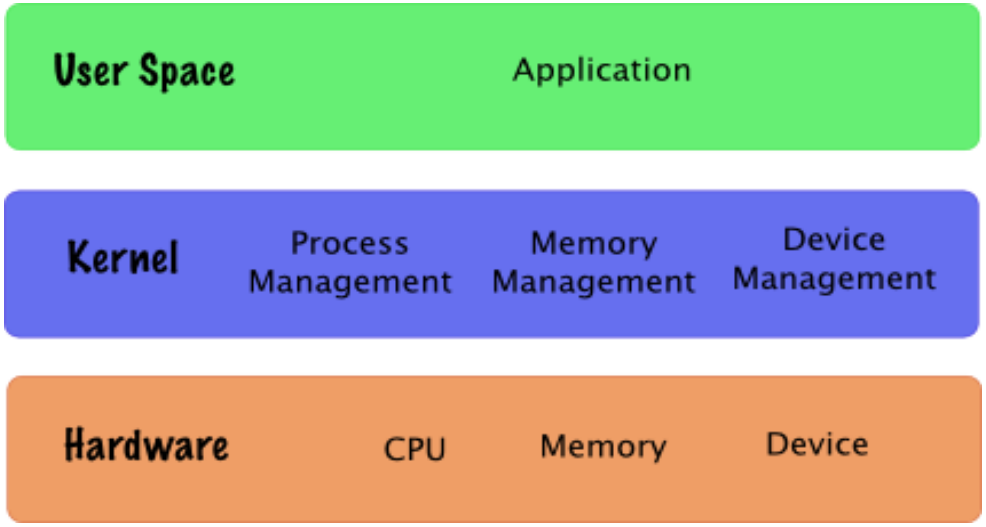
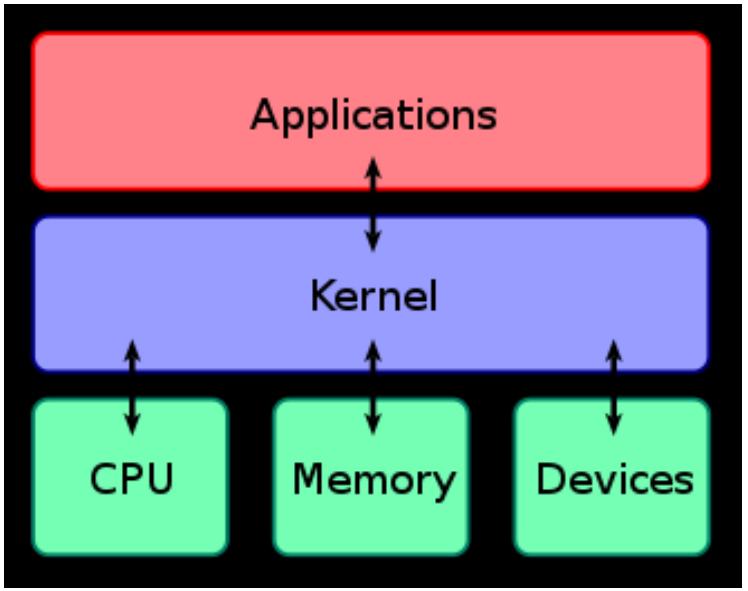


Figure 2.2 The Operating System as Resource Manager

Kernel

- Portion of operating system that is in main memory
- Contains most frequently used functions



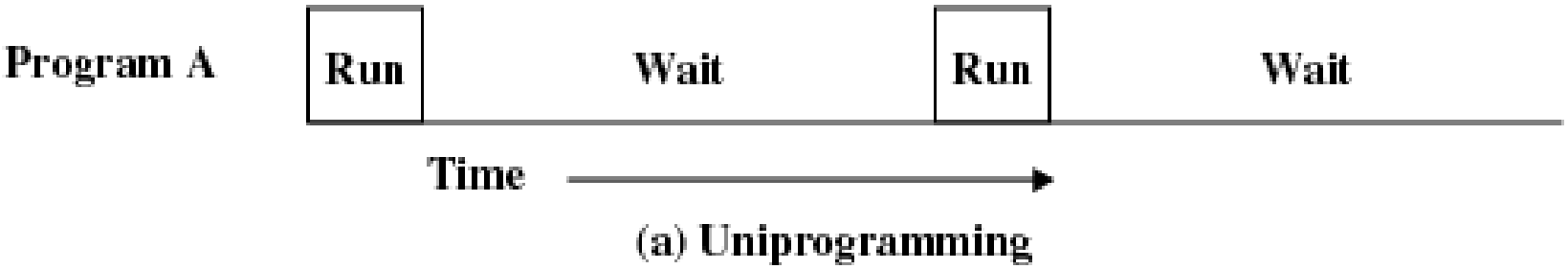


Kernel vs. User

- https://www.youtube.com/watch?v=o_JCGR_9HRs

Uniprogramming

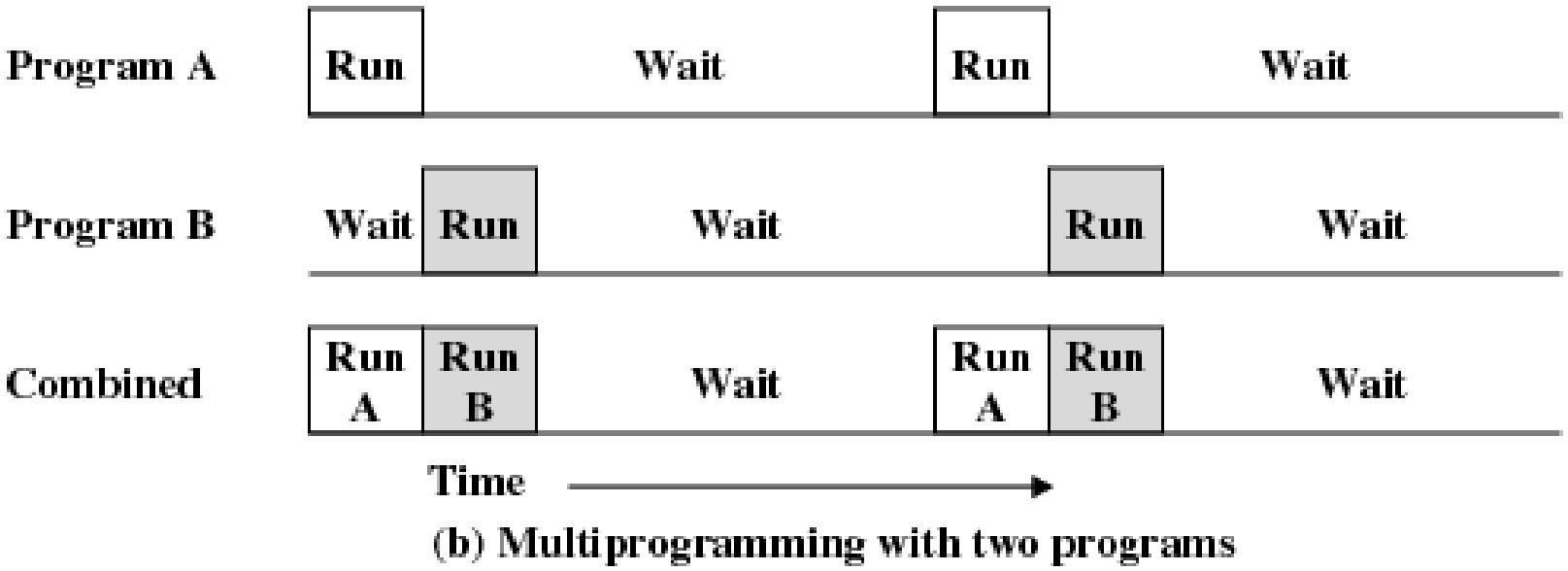
- Processor must wait for I/O instruction to complete before preceding



(a) Uniprogramming

Multiprogramming

- When one job needs to wait for I/O, the processor can switch to the other job



(b) Multiprogramming with two programs

Timing

Prevents a job from monopolizing the system

Process	Arrival time	Burst time
P1	0 ms	10 ms
P2	0 ms	5 ms
P3	0 ms	8 ms

Gantt Chart

P1		P2		P3		P1		P2		P3	
1	2	2	4	4	6	6	8	8	10	10	12

P1		P2		P3		P1		P3		P1	
12	14	14	15	15	17	17	19	19	21	21	23

AfterAcademy

Utilization Histograms

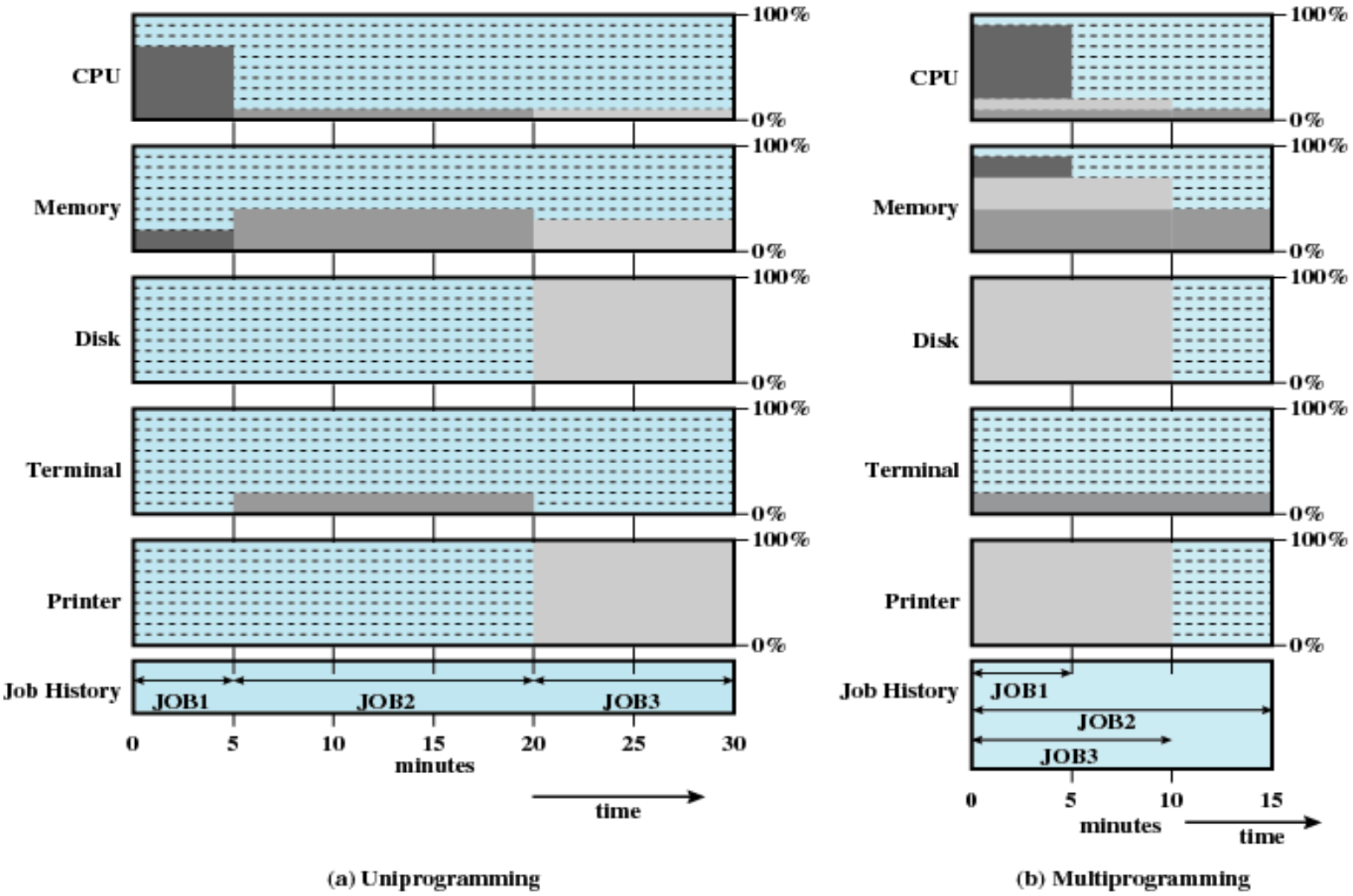


Figure 2.6 Utilization Histograms

I/O Devices Slow

Read one record from file	15 μs
Execute 100 instructions	1 μs
Write one record to file	15 μs
TOTAL	31 μs

$$\text{Percent CPU Utilization} = \frac{1}{31} = 0.032 = 3.2\%$$

Figure 2.4 System Utilization Example

Multitasking

- Idea:
  - More than one process can be running at once
  - When one process blocks (waiting for disk, network, user input, etc.) run another process
- Problem:
  - What can ill-behaved process do?
  - Go into infinite loop and never relinquish CPU- Scribble over other processes' memory to make them fail
- OS provides mechanisms to address these problems
  - Preemption–take CPU away from looping process
  - Memory protection–protect processes' memory from one another

Key Elements of Operating System

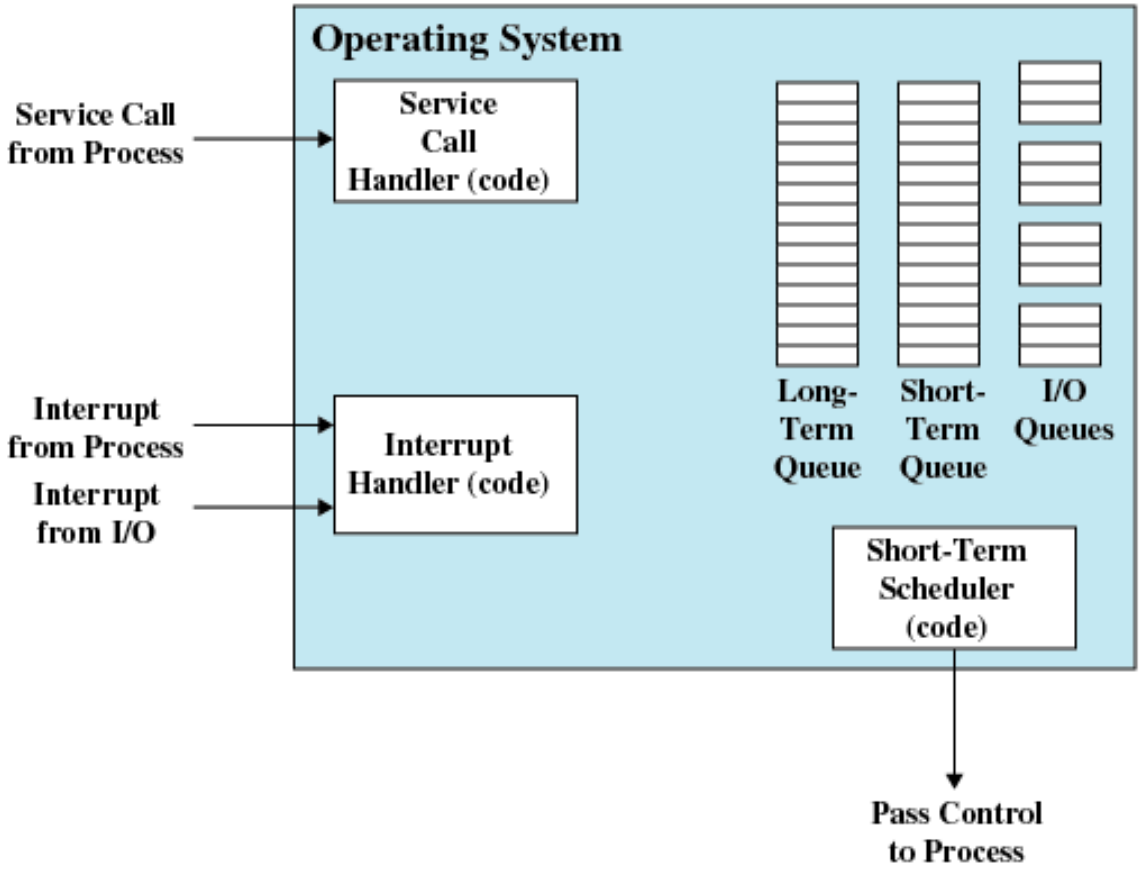


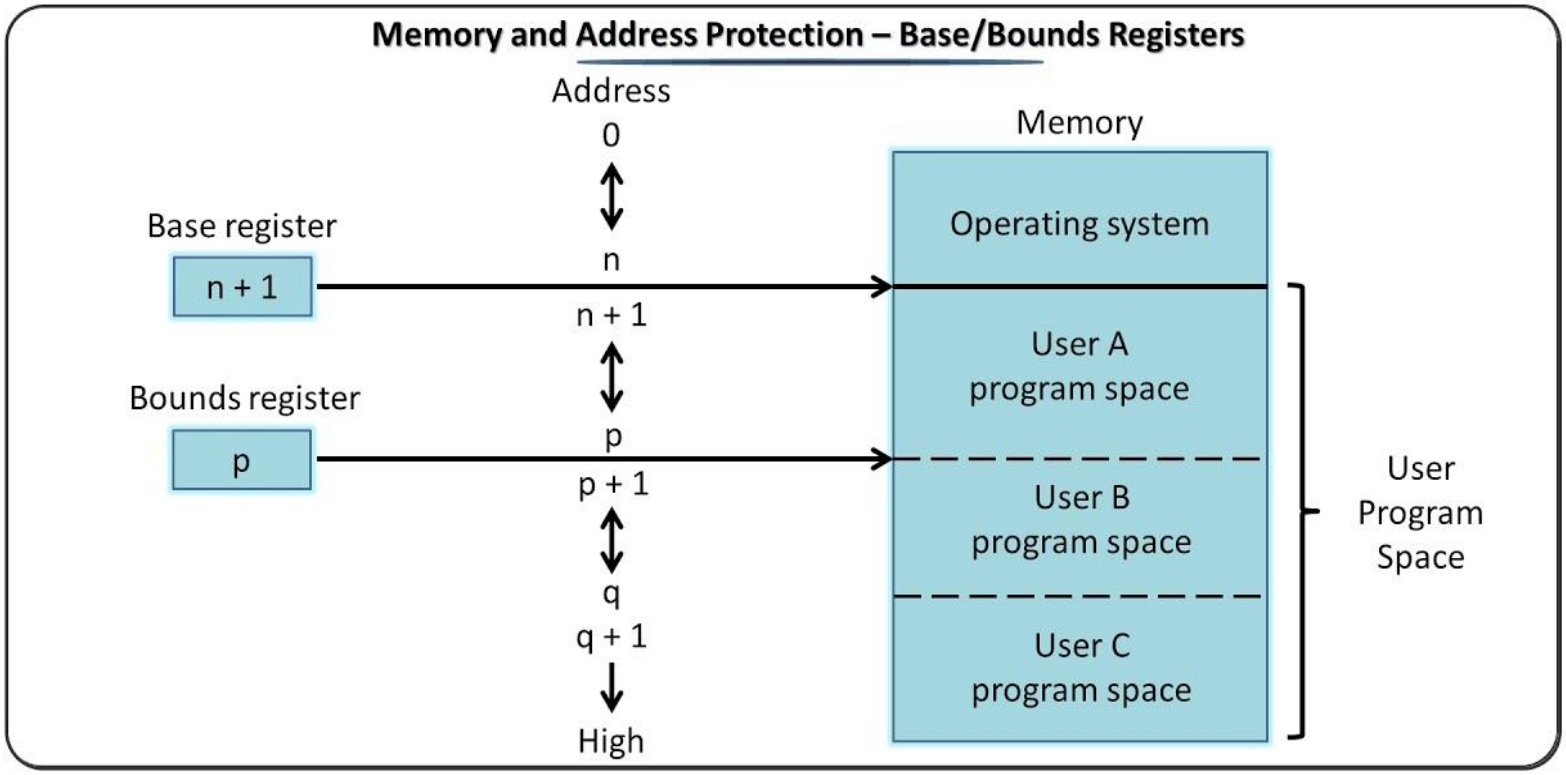
Figure 2.11 Key Elements of an Operating System for Multiprogramming

Protection

- User program executes in user mode
  - Certain instructions may not be executed
- Monitor executes in system mode
  - Kernel mode
  - Privileged instructions are executed
  - Protected areas of memory may be accessed

Memory protection

Do not allow the memory area containing the monitor to be altered

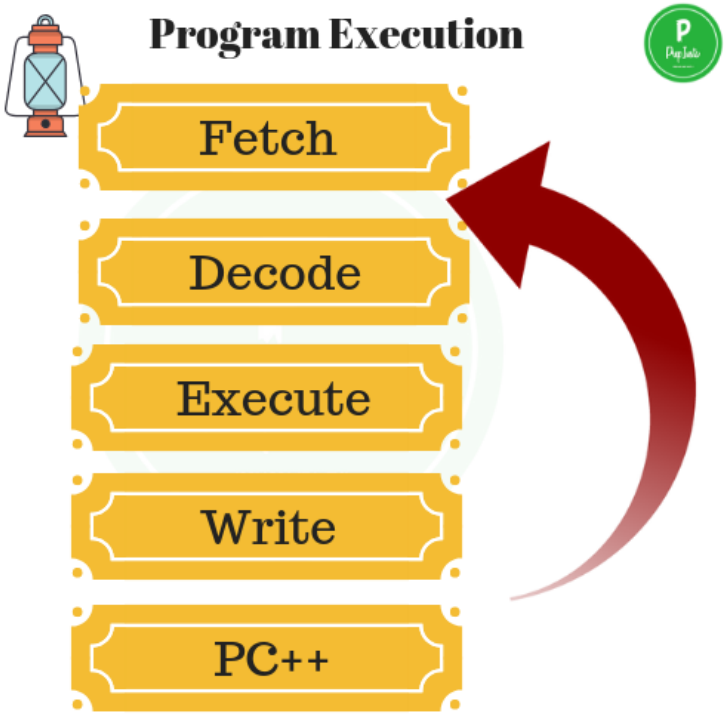


Scheduling and Resource Management

- Fairness
  - Give equal and fair access to resources
- Differential responsiveness
  - Discriminate among different classes of jobs
- Efficiency
  - Maximize throughput, minimize response time, and accommodate as many uses as possible

Processes (1/2)

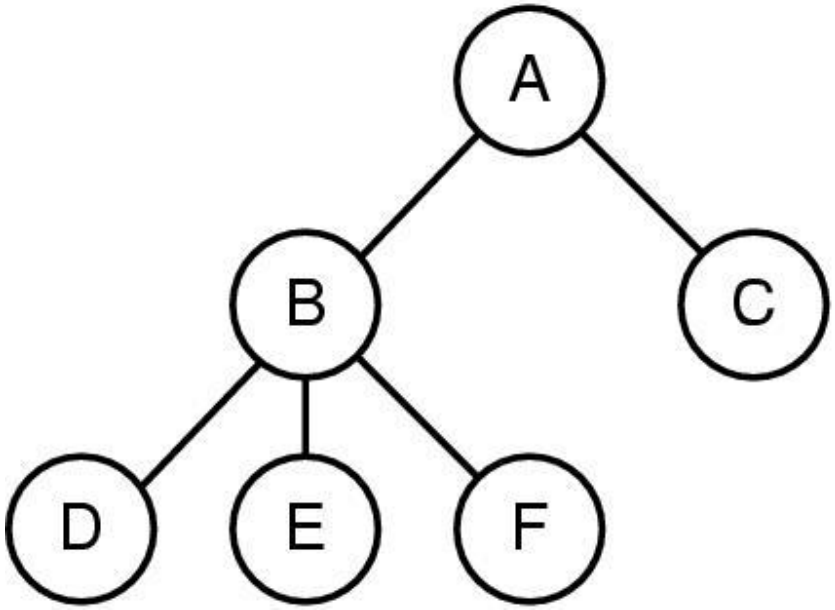
- A program in execution
- An instance of a program running on a computer
- The entity that can be assigned to and executed on a processor
- A unit of activity characterized by a single sequential thread of execution, a current state, and an associated set of system resources



Process (2/2)

- Consists of three components
  - An executable program
  - Associated data needed by the program
  - Execution context of the program
    - All information the operating system needs to manage the process

Operating System Concepts (1)



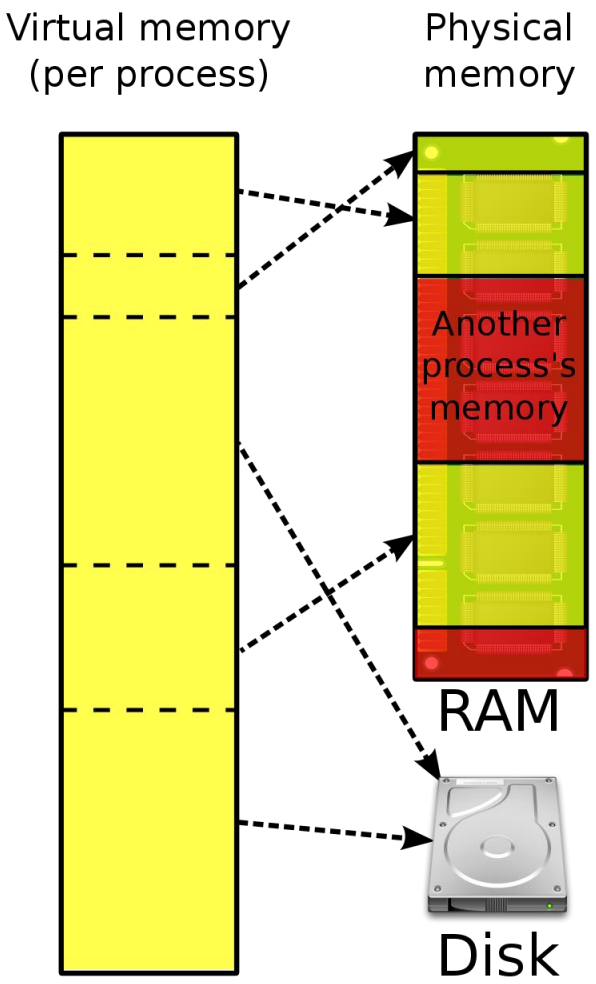
- A process tree
  - A created two child processes, B and C
  - B created three child processes, D, E, and F

Memory Management

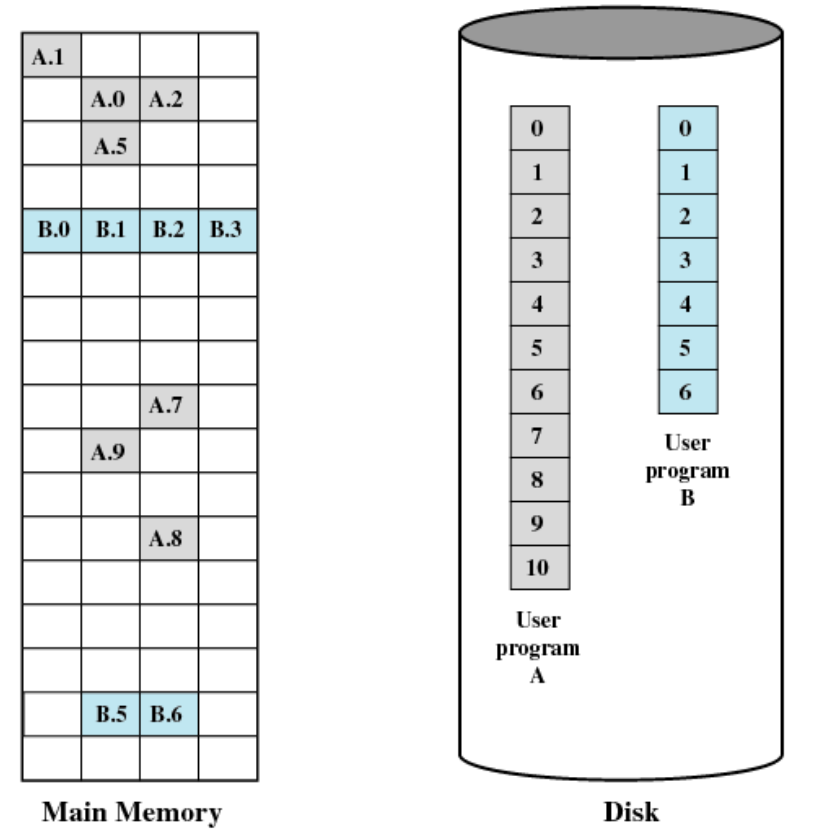
- Process isolation
- Automatic allocation and management
- Support of modular programming
- Protection and access control
- Long-term storage

Virtual Memory and File System

- Implements long-term store
- Information stored in named objects called files



Virtual Memory



Main memory consists of a number of fixed-length frames, each equal to the size of a page. For a program to execute, some or all of its pages must be in main memory.

Secondary memory (disk) can hold many fixed-length pages. A user program consists of some number of pages. Pages for all programs plus the operating system are on disk, as are files.

Figure 2.9 Virtual Memory Concepts

Virtual Memory Addressing

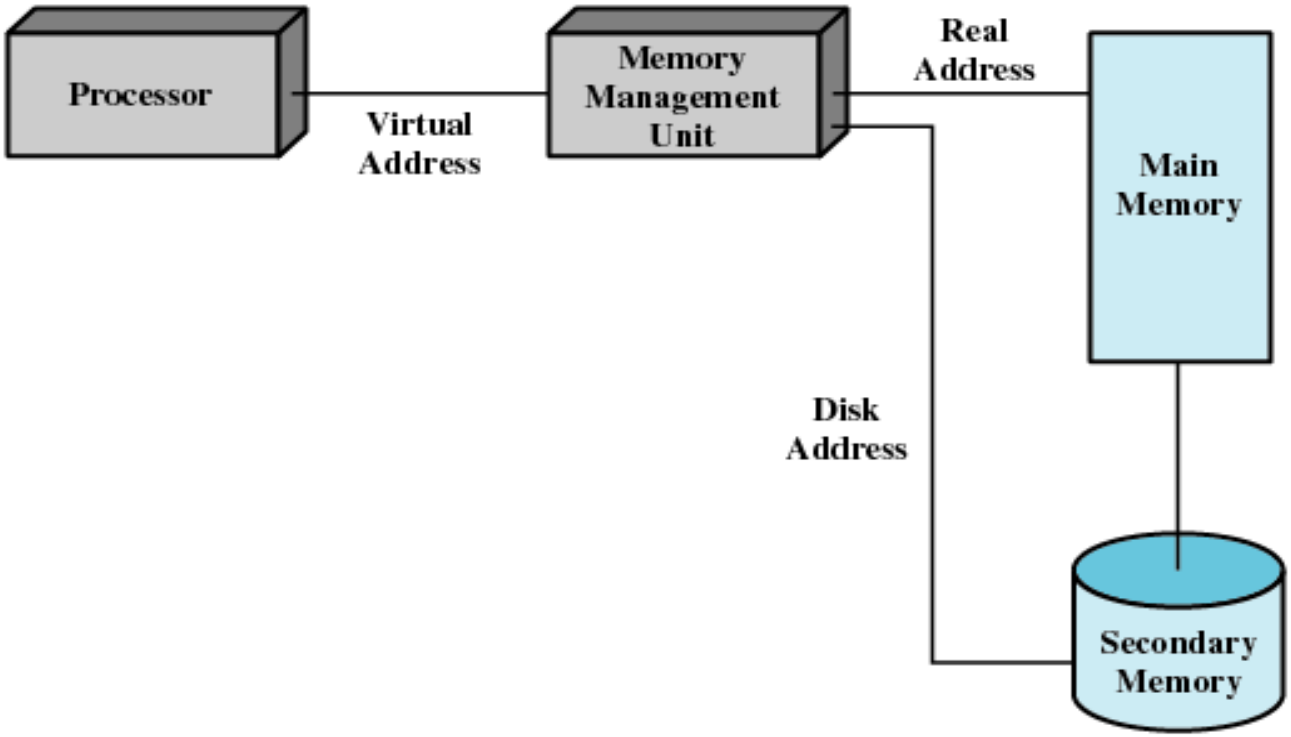


Figure 2.10 Virtual Memory Addressing

Information Protection and Security

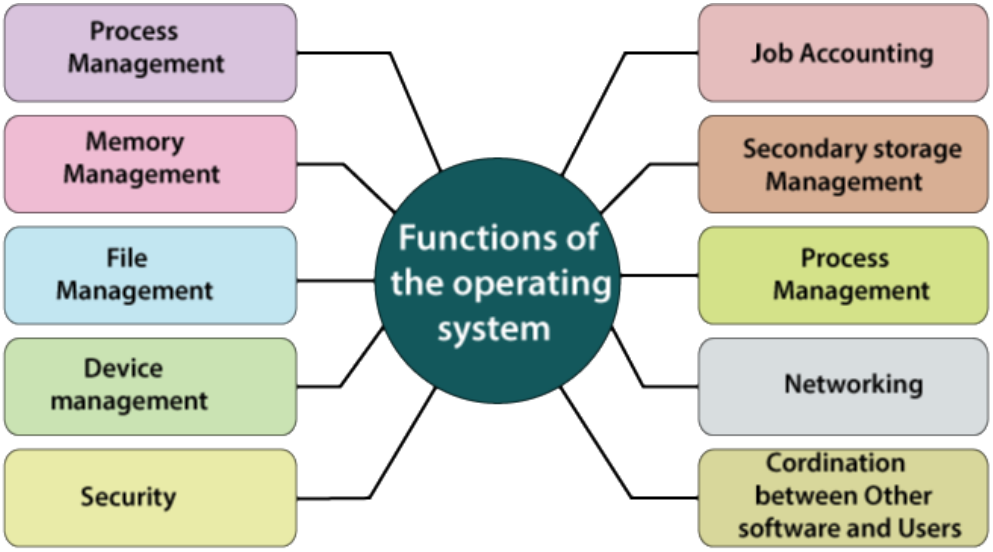
- Availability
  - Concerned with protecting the system against interruption
- Confidentiality
  - Assuring that users cannot read data for which access is unauthorized

Information Protection and Security

- Data integrity
  - Protection of data from unauthorized modification
- Authenticity
  - Concerned with the proper verification of the identity of users and the validity of messages or data

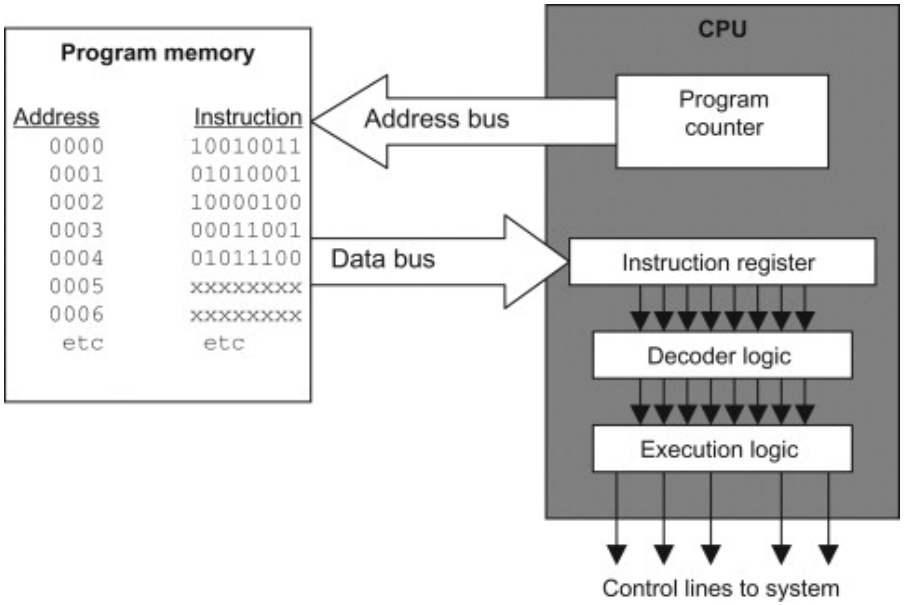
System Structure

- View the system as a series of levels
- Each level performs a related subset of functions
- Each level relies on the next lower level to perform more primitive functions
- This decomposes a problem into a number of more manageable subproblems



Process Hardware Levels

- Level 1
  - Electronic circuits
  - Objects are registers, memory cells, and logic gates
  - Operations are clearing a register or reading a memory location



Process Hardware Levels

- Level 2
  - Processor's instruction set
  - Operations such as add, subtract, load, and store
- Level 3
  - Adds the concept of a procedure or subroutine, plus call/return operations
- Level 4
  - Interrupts

Concepts with Multiprogramming

- Level 5
  - Process as a program in execution
  - Suspend and resume processes
- Level 6
  - Secondary storage devices
  - Transfer of blocks of data
- Level 7
  - Creates logical address space for processes
  - Organizes virtual address space into blocks

Deal with External Objects

- Level 8
  - Communication of information and messages between processes
- Level 9
  - Supports long-term storage of named files
- Level 10
  - Provides access to external devices using standardized interfaces

Deal with External Objects

- Level 11
  - Responsible for maintaining the association between the external and internal identifiers
- Level 12
  - Provides full-featured facility for the support of processes
- Level 13
  - Provides an interface to the operating system for the user

- Microkernel architecture
  - Assigns only a few essential functions to the kernel
    - Address spaces
    - Interprocess communication (IPC)
    - Basic scheduling

- Multithreading
  - Process is divided into threads that can run concurrently
    - Thread
      - Dispatchable unit of work
      - executes sequentially and is interruptable
    - Process is a collection of one or more threads

- Symmetric multiprocessing (SMP)
  - There are multiple processors
  - These processors share same main memory and I/O facilities
  - All processors can perform the same functions

Multiprogramming and Multiprocessing

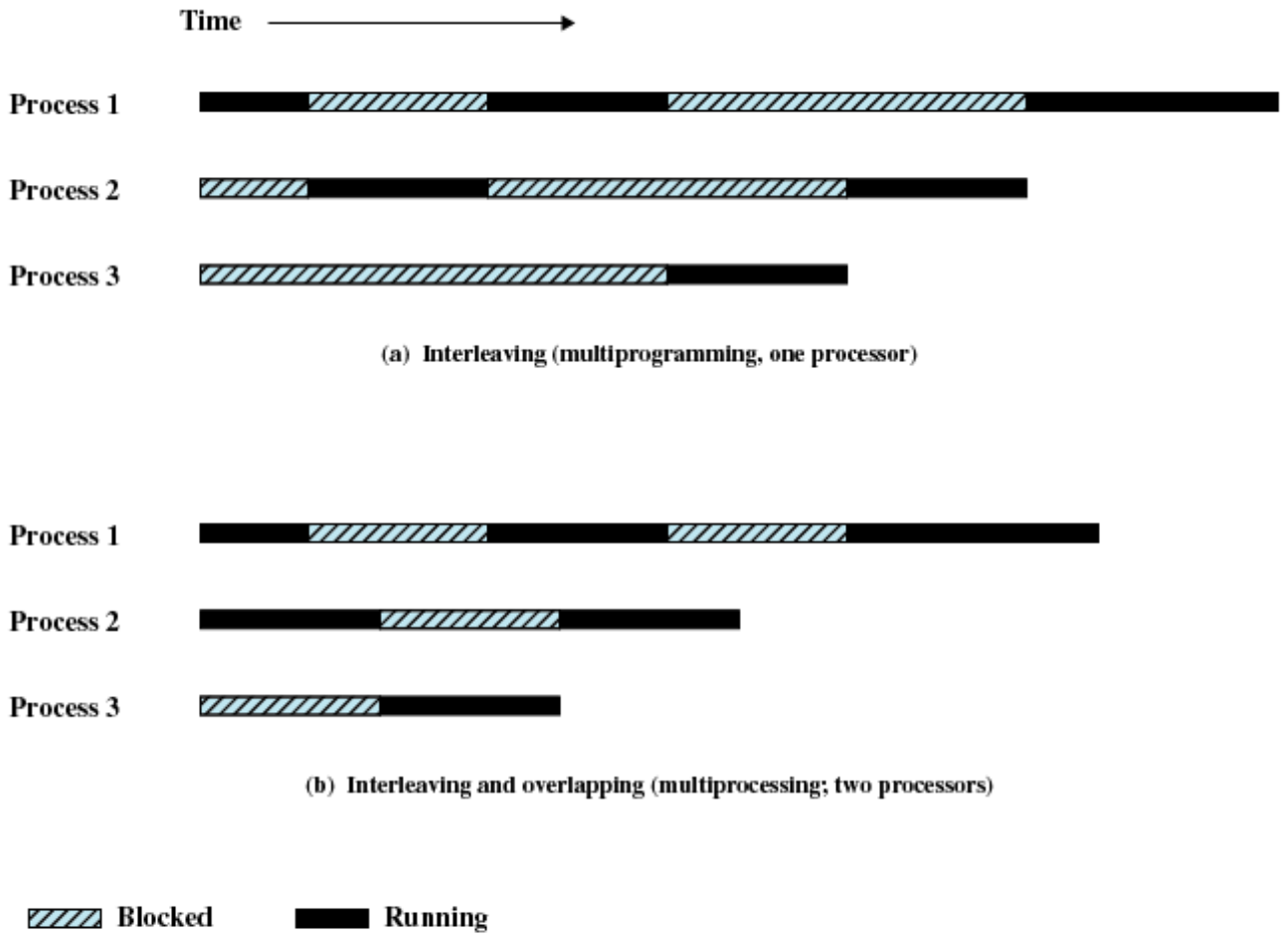


Figure 2.12 Multiprogramming and Multiprocessing

Modern Operating Systems

- Distributed operating systems
  - Provides the illusion of a single main memory space and single secondary memory space

- Object-oriented design
  - Used for adding modular extensions to a small kernel
  - Enables programmers to customize an operating system without disrupting system integrity

Kernel-Mode Components

- Executives
  - Contains base operating system services
    - Memory management
    - Process and thread management
    - Security
    - I/O
    - Interprocess communication
- Kernel
  - Consists of the most used components

Kernel-Mode Components

- Hardware abstraction layer (HAL)
  - Isolates the operating system from platform-specific hardware differences
- Device drivers
  - Translate user I/O function calls into specific hardware device I/O requests
- Windowing and graphics systems
  - Implements the graphical user interface (GUI)

User-Mode Processes

- Special system support processes
  - Ex: logon process and the session manager
- Service processes
- Environment subsystems
- User applications

Client/Server Model

- Simplifies the Executive
  - Possible to construct a variety of APIs
- Improves reliability
  - Each service runs on a separate process with its own partition of memory
  - Clients cannot not directly access hardware
- Provides a uniform means for applications to communicate via local procedure call (LPC)
- Provides base for distributed computing

UNIX

- Hardware is surrounded by the operating system software
- Operating system is called the system kernel
- Comes with a number of user services and interfaces
  - Shell
  - Components of the C compiler

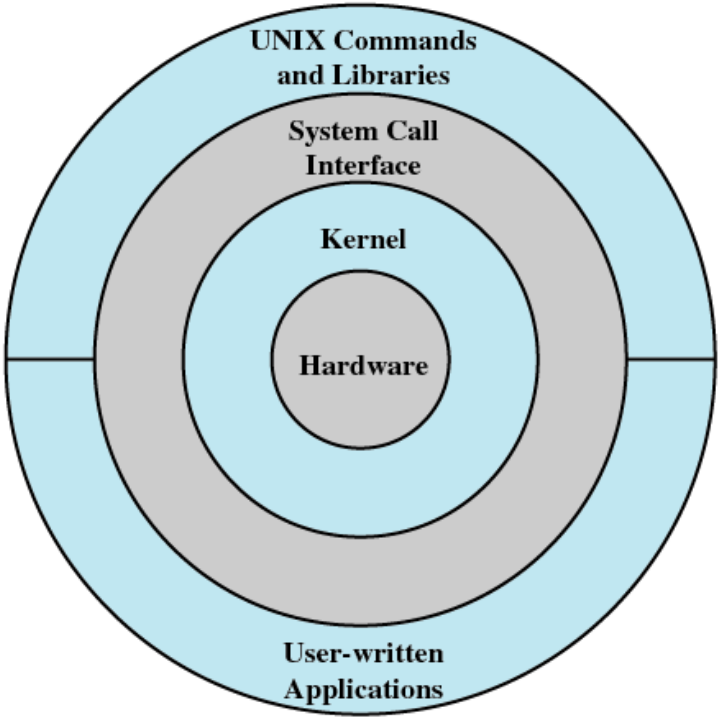


Figure 2.14 General UNIX Architecture

UNIX Kernel

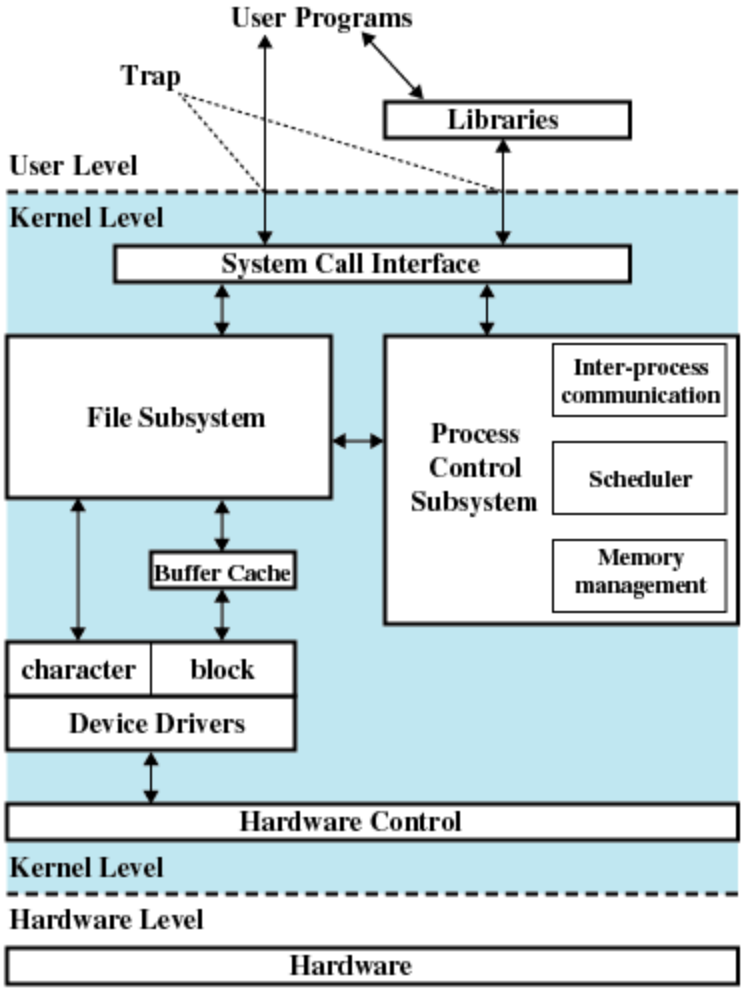
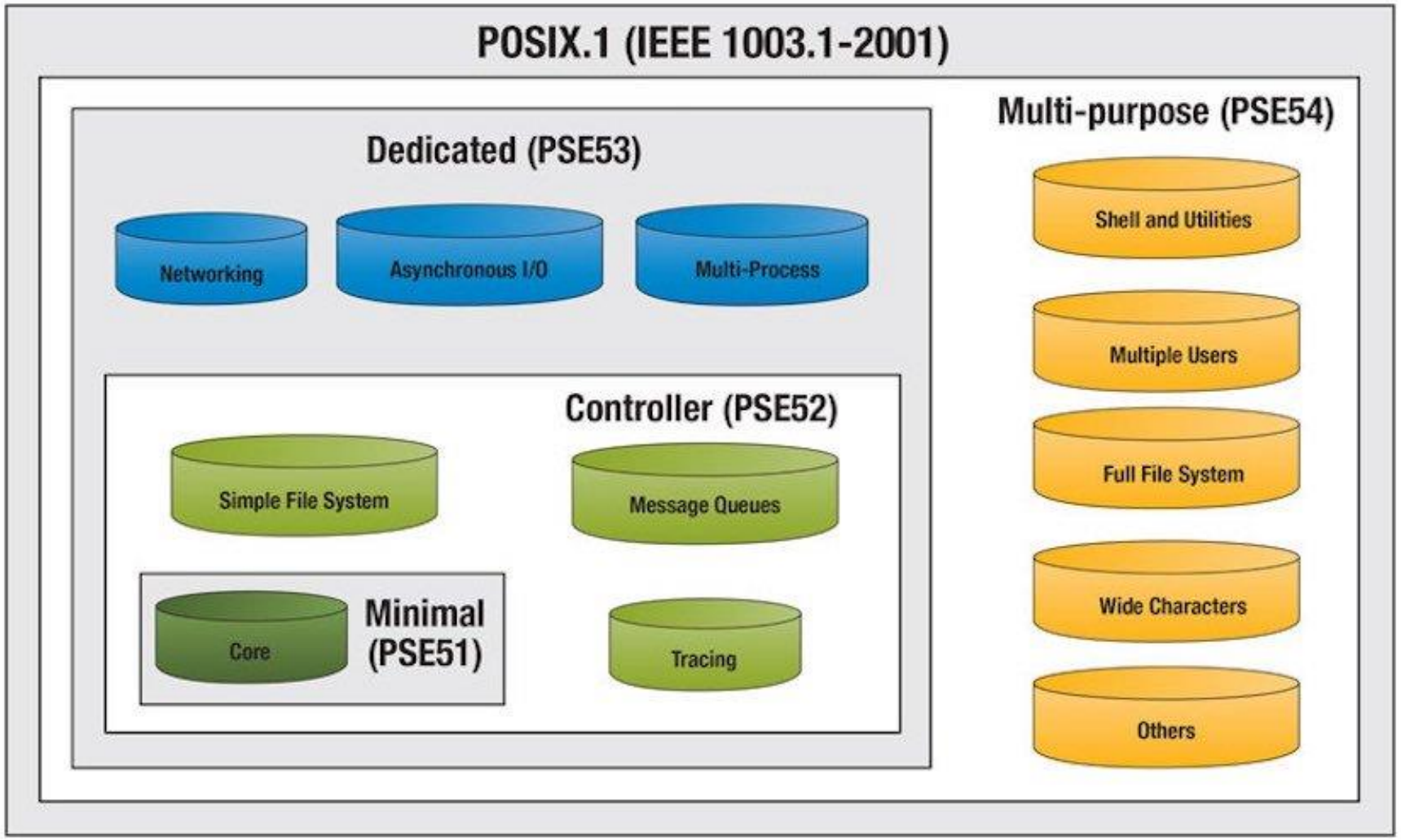


Figure 2.15 Traditional UNIX Kernel [BACH86]

POSIX (Portable Operating System Interface)



POSIX (Portable Operating System Interface)

Fully certified	Mostly complied	MS Windows supplements
HP UX	Linux	Cywin
Solaris	Android	MinGW
AIX	BSD	Windows Subsystem for Linux
QNX	Nucleus RTOS	Microsoft POSIX subsystem

POSIX subsystem implementation.

